

# INFEROPT: Constrained Multi-Objective Search for LLM Inference Configurations

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## Abstract

Serving an LLM means setting dozens of inference-time knobs, from per-layer KV retention to per-layer expert counts. Practice sets them with mechanism-specific heuristics that return a single operating point and do not scale to layer-wise search spaces. We recast inference configuration as constrained multi-objective black-box optimization and build INFEROPT, a reusable search framework that requires only variable bounds, a deterministic resource cost, and an evaluation hook. INFEROPT searches on a frozen sampled proxy set, rejects over-budget candidates before any model call, tightens the budget adaptively, and re-validates Pareto representatives on full-scale dataset. One pipeline covers a 28-dimensional continuous KV space (Qwen2.5-7B) and a 26-dimensional discrete MoE space (DeepSeek-V2-Lite). On KV, post-prefill pruning cuts the 16K cache by 64.4% and TPOT by 22.9–48.5%, and the searched layer-wise budget by INFEROPT beats a matched uniform budget by 7.3% and 14.0% of the Full KV reference points. On MoE, a searched top- $k$  schedule by INFEROPT removes 43.0% of routed token–expert pairs while staying within 0.59 points of the default, closer than the matched-budget baselines. Against Random Search, NSGA-II, and MOTPE, INFEROPT leads on both spaces, taking the best proxy hypervolume and the lowest retention on KV and staying closest to the uncompressed reference at the lowest experts on MoE.

## 1 Introduction

LLM serving systems expose configuration choices for KV-cache compression [11], sparse attention, MoE routing [12], and quantization, which are typically determined by uniform policies or mechanism-specific importance heuristics [8]. With dozens of layer-wise variables, these heuristics become inadequate. They explore the high-dimensional configuration space only narrowly and yield a single operating point rather than the quality-efficient Pareto frontier required under varying deployment budgets.

We argue the missing piece is not another compression heuristic but an *optimization layer*. InferOpt represents continuous KV-memory fractions and discrete routing decisions within a unified mixed-domain configuration space. Supporting a new mechanism requires only the specification of its variable domains, resource-cost function, and evaluation interface. The same search and validation pipeline is reused without mechanism-specific redesign. Evaluation cost is the primary bottleneck. Evaluating a single configuration of a billion-parameter model on the full benchmark can take hours, making conventional multi-objective search impractical under realistic budgets and motivating a sample-efficient search framework.

INFEROPT addresses these challenges through three empirically validated components. **(i) Frozen proxy evaluation with explicit reliability auditing.** A fixed proxy set ranks candidate configurations during search, after which selected configurations undergo full-benchmark validation. **(ii) Deterministic resource-aware pruning.** A deterministic resource-cost function rejects over-budget candidates

before model evaluation and progressively tightens the admissible budget as feasible observations accumulate. **(iii) A mechanism-agnostic search interface.** By exposing only variables, constraints, and objectives, the same search and validation pipeline supports both continuous KV-cache and discrete MoE-routing spaces without mechanism-specific redesign. Across both spaces, INFEROPT outperforms Random Search, NSGA-II, and MOTPE.

**Related work.** Existing inference-acceleration methods [11, 10, 18] introduce effective configuration mechanisms, but typically optimize them in isolation. General-purpose multi-objective optimization methods [3, 1, 14] support heterogeneous search spaces, but rely on relatively inexpensive objective evaluations. INFEROPT bridges this gap by extending the abstraction of ML auto-tuners [4, 17] from low-level tensor schedules to expensive, high-level inference configurations.

## 2 Method

Let  $\mathbf{x} \in \mathcal{X}$  collect per-layer retention ratios, expert counts, or any other inference configuration. We consider the following multi-objective optimization problem including resource cost and quality loss,

$$\min_{\mathbf{x} \in \mathcal{X}} (\text{ResourceCost}(\mathbf{x}), Q_{\text{ref}} - Q(\mathbf{x}, \mathcal{D})) \quad \text{s.t.} \quad g_j(\mathbf{x}) \leq 0, \quad (1)$$

where  $Q_{\text{ref}}$  is the score of the uncompressed reference on the same examples. Resource cost is an architecture-level proxy computed in closed form, namely mean retained historical KV blocks or mean activated routed experts, not measured latency, memory, or energy.

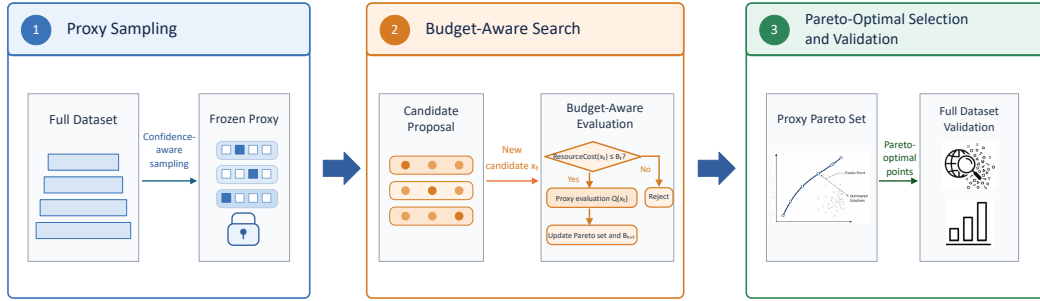


Figure 1: INFEROPT in three stages. Proxy sampling constructs a frozen stratified subset; budget-aware multi-objective search rejects over-budget candidates before evaluation; Pareto-optimal representatives are selected and re-validated on disjoint held-out examples.

Figure 1 illustrates the three-stage INFEROPT pipeline. **Proxy sampling.** We first construct a proxy subset from the full dataset using confidence-aware sampling and freeze it before search. Consequently, every candidate is evaluated on the same examples. **Budget-aware search.** At iteration  $t$ , multi-objective TPE proposes a continuous or discrete configuration  $\mathbf{x}_t$  based on the Pareto structure of previously evaluated trials [3]. Before model evaluation, a closed-form resource-cost function checks whether  $\text{ResourceCost}(\mathbf{x}_t) \leq B_t$ . Over-budget candidates are rejected immediately, whereas feasible candidates are evaluated on the frozen proxy set and used to update both the proxy Pareto set and the next-round budget  $B_{t+1}$ . **Pareto-optimal selection and validation.** Finally, Pareto-optimal configurations are selected from the proxy Pareto set and re-evaluated on the full dataset to identify deployment-ready operating points.

**Two instantiations.** We instantiate INFEROPT on two structurally different configuration spaces. **KV-cache compression.** The configuration is a 28-dimensional continuous vector  $\boldsymbol{\rho} \in [0.1, 1.0]^{28}$ , where  $\rho_\ell$  denotes the retention ratio of layer  $\ell$ . We define the resource cost as the average retention ratio,  $\bar{\rho} = \frac{1}{28} \sum_{\ell=1}^{28} \rho_\ell$ , and impose the adaptive constraint  $\bar{\rho} \leq B_t$ , where  $B_0 = 1.0$  and  $B_t$  is lower-bounded by 0.2. **MoE routing.** The configuration is a 26-dimensional discrete vector  $\mathbf{k} \in \{1, \dots, 6\}^{26}$ , where  $k_\ell$  denotes the number of activated experts at layer  $\ell$ . Its resource cost is  $\bar{k} = \frac{1}{26} \sum_{\ell=1}^{26} k_\ell$ , with no hard constraint imposed on the total number of activated experts during search. Apart from the variable domains and resource-cost definitions, the search and validation pipeline remains unchanged across the two instantiations.

### 3 Experiments

**Setup.** All runs use one GPU, Python 3.11, PyTorch 2.5.1, and bfloat16. KV compression uses Qwen2.5-7B-Instruct [13] on LongBench [2], with 210 proxy and 1,050 held-out examples over 21 tasks and a protected 128-token prefix and 512-token recent window. MoE routing uses DeepSeek-V2-Lite-Chat [7] with 64 experts per routed layer, on 384 proxy and 768 held-out MMLU/GSM8K/BBH examples [9, 5, 15] and checkpoint default  $k_\ell = 6$ . Multi-objective search baselines are Random Search, NSGA-II [6], and MOTPE [3, 1] under the same box constraints but without adaptive budget-tightening. Appendix A gives the remaining protocol details.

Table 1: Comparison of multi-objective search methods. Left: KV proxy search over 24 trials, hypervolume against the reference point on the 210-example proxy. The last column is quality loss in points at the minimum-retention trial, where full KV scores 40.25/100. Right: MoE 24-trial replay on the 384-example proxy, max and mean score comes from six selected candidates per method. The uncompressed default  $k_\ell = 6$  reference scores 0.1953 (19.53/100). MoE scores are read by proximity to that reference rather than by magnitude. Bold marks the best value in a column.

| Method   | Hypervolume    | Min. $\bar{\rho}$ | Loss at min. $\bar{\rho}$ | Method   | Max           | Mean          | $\sum_\ell k_\ell$ | $\bar{k}$ |
|----------|----------------|-------------------|---------------------------|----------|---------------|---------------|--------------------|-----------|
| Random   | 0.00788        | 0.313             | 0.166                     | Random   | 0.2240        | 0.1836        | 65.7               | 2.53      |
| NSGA-II  | 0.00795        | 0.331             | -0.033                    | NSGA-II  | 0.2179        | 0.1832        | 95.3               | 3.67      |
| MOTPE    | 0.00778        | 0.327             | 0.005                     | MOTPE    | 0.2240        | 0.1840        | 75.7               | 2.91      |
| INFEROPT | <b>0.00881</b> | <b>0.274</b>      | 0.053                     | INFEROPT | <b>0.1931</b> | <b>0.1910</b> | <b>59.3</b>        | 2.28      |

**Search-method comparison.** Table 1 compares the four multi-objective search methods. On the KV proxy, INFEROPT attains the highest mean hypervolume (0.00881, 10.8% above the best baseline) and the lowest minimum retention (0.274, 12.5% below it) at a quality loss of 0.053 points (0.1% of the Full KV proxy reference). On the MoE panel, INFEROPT leads all three columns. Its top candidate scores 0.1931, within 0.22 points of the reference (1.1%), against 11.6–14.7% for the three baselines; its selected set averages 0.1910, again the closest at 2.2%; and it reaches both at the lowest budget,  $\sum_\ell k_\ell = 59.3$ , 9.7–37.8% fewer activated experts than the baselines.

#### 3.1 KV cache compression

**Controlled comparison.** We fix an aggressive mean-retention budget ( $\bar{\rho} = 0.321$ ) and evict unselected KV blocks immediately after prefill. We independently vary *allocation* and *selection*: allocation uses either the layer-wise vector found by INFEROPT or a matched uniform vector, while selection uses attention-mass scoring [10], key- $\ell_2$  magnitude, or an H2O-style policy [16]. All five compressed policies share the same protected regions and mean budget, enabling controlled comparisons of the allocation and selection effects.

**Quality.** Figure 2(a) shows that all compressed policies trade quality for efficiency, decreasing the Full-KV score of 43.83 by 5.79–10.78 points on 1,050 validation examples. The comparisons show that allocation has the larger effect. With allocation fixed, attention-mass scoring outperforms key- $\ell_2$  by only 0.52 points. With the selector fixed, the searched allocation outperforms its budget-matched uniform counterpart by 3.21 points with attention scoring and 6.13 points with key- $\ell_2$ .

**Efficiency and search behavior.** At a 16K context length, panel (b) shows that INFEROPT reduces KV-cache memory from 902.9 to 319.1–321.5 MiB (64.4%) and TPOT from 19.11 ms to 9.83 ms with attention scoring and 10.22 ms with key- $\ell_2$ . These latencies are lower than those of the matched uniform policies (13.34–13.36 ms) and the H2O-style baseline (14.74 ms). Panel (c) visualizes the underlying search: each orange point denotes a candidate evaluated, and the black curve traces the resulting proxy Pareto front. At nearly identical cache budgets, the selected Pareto-frontal configurations outperform the matched uniform baselines by 3.21 and 6.13 (7.3% and 14.0%) quality points and reduce TPOT by 3.5 and 3.1 ms (26.4% and 23.4%), respectively. Thus, INFEROPT exposes the full quality–efficiency frontier.

#### 3.2 MoE top- $k$ routing

Table 2 compares five layer-wise top- $k$  schedules on a 768-example validation set: the INFEROPT schedule, the uncompressed default, two budget-matched controls, and a lower-budget uniform

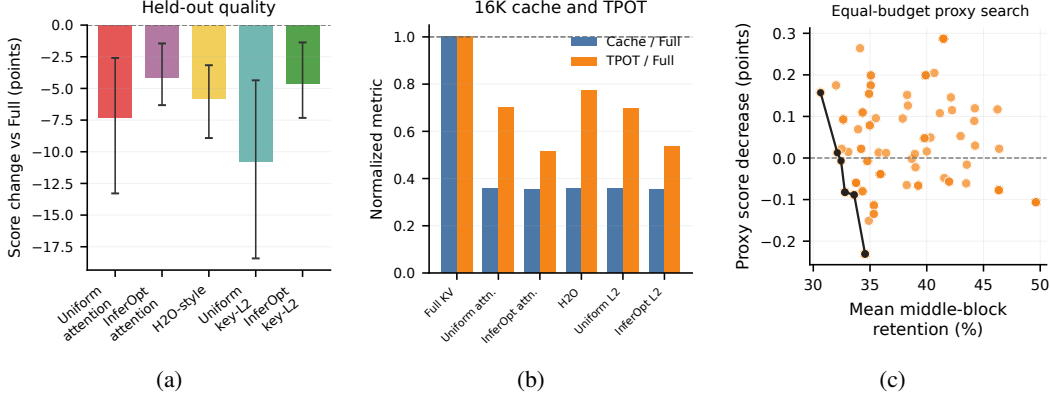


Figure 2: KV results at a fixed aggressive budget  $\bar{\rho} = 0.321$ . (a) Validation/Held-out quality change against Full KV on 1,050 paired examples; Full KV scores 43.83/100. (b) Same-process 16K cache and TPOT, normalized to Full KV over ten trials. (c) Every KV compression candidate evaluated during the proxy search of InferOpt, black curve marking the non-dominated front.

endpoint. Routed pairs counts token-expert assignments, and TPOT is measured at an 8K context length. Score denotes macro-averaged validation quality, while  $\Delta$  measures its absolute deviation from the uncompressed default. The selected schedule, `seg_early6_mid3_late1`, assigns  $k = 6, 3, 1$  to the first 9, middle 9, and final 8 layers, respectively, yielding  $\bar{k} = 3.42$ . It achieves the smallest quality deviation, remaining within 0.59 points (3.3%) of the default, while reducing routed pairs by 43.0% and TPOT by 19.3%. These results indicate that preserving expert capacity in early layers while aggressively pruning later layers provides a favorable quality-efficiency trade-off.

Two matched-budget controls test that shape at  $\sum_{\ell} k_{\ell} = 89$ . Near-uniform spreads the 11 extra experts over evenly spaced layers; entropy-based allocation gives them to the 11 layers with the highest normalized entropy of default- $k$  activation counts on the proxy. The segmented schedule lands 0.45 points (2.5% of the default score) closer to the default than near-uniform and 0.97 points (5.4%) closer than entropy-based. We built both controls only after the winning schedule was known, so they are a post-hoc check rather than a pre-registered test. The three schedules spend the same budget and differ only in which layers get the extra experts, so the gap between them cannot come from budget, and their TPOT differences are runtime effects for the same reason.

Table 2: MoE quality and 8K system measurements over full validation set. The two matched-budget controls are post-hoc; uniform  $k = 3$  is a lower-budget endpoint rather than a matched control.

| Configuration                       | Mean $k$ | Score  | $\Delta$ | Routed pairs | TPOT (ms) |
|-------------------------------------|----------|--------|----------|--------------|-----------|
| <code>seg_early6_mid3_late1</code>  | 3.42     | 0.1725 | -0.0059  | 17,406       | 32.46     |
| Default $k = 6$                     | 6.00     | 0.1784 | 0.0000   | 30,510       | 40.23     |
| Near-uniform, $\sum k_{\ell} = 89$  | 3.42     | 0.1680 | -0.0104  | 17,406       | 35.73     |
| Entropy-based, $\sum k_{\ell} = 89$ | 3.42     | 0.1628 | -0.0156  | 17,406       | 35.72     |
| Uniform $k = 3$                     | 3.00     | 0.1589 | -0.0195  | 15,255       | 32.26     |

## 4 Conclusion

We introduced INFEROPT, a unified constrained multi-objective search framework in which each inference mechanism specifies only its variable domains, resource-cost function, and evaluation interface. The same pipeline supports continuous KV-retention and discrete MoE-routing spaces without mechanism-specific redesign and outperforms Random Search, NSGA-II, and MOTPE in both settings. The searched KV allocation improves quality by 3.21–6.13 points (7.3% and 14.0%) and reduces TPOT by 3.1–3.5 ms (26.4% and 23.4%) over uniform allocation, while using 64.4% less cache than Full KV at 16K. The searched MoE schedule reduces routed work by 43.0% while remaining within 0.59 points of the default validation score. These results demonstrate that a shared, budget-aware search framework can effectively replace mechanism-specific tuning heuristics.

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## A Protocol details

**KV compression.** Every compressed policy uses `prefill_after` compression with a 128-token protected prefix, a 512-token recent window, 128 middle blocks, a 32-step compression interval, and 128-token prefill attention query chunks, at a fixed mean middle-block retention  $\bar{\rho} = 0.321$ . Compressed policies prune once immediately after prefill and then at the compression interval during decoding, while Full KV is a generation-only baseline. Held-out quality is the macro-average over 21 LongBench tasks on 1,050 paired examples, with intervals from 1,000 paired bootstrap resamples. The system matrix covers six strategies at 4K, 8K, and 16K context with one warm-up run, ten measured repetitions, and 128 generated tokens per run; the main text reports the 16K endpoint. Because the five compressed policies share protected regions and mean budget, uniform attention, uniform key-L2, and H2O-style retain the same block count and the two INFEROPT policies share one layer-wise vector; the 321.5 versus 319.1 MiB gap is per-layer block rounding.

**MoE routing.** The proxy objective is the macro-average over the 384-example MMLU/GSM8K/BBH split; a 0.02 quality tolerance labels proxy candidates but is not a hard feasibility constraint. The proxy-to-held-out Kendall correlation is  $\tau_b = 0.59$  with 80% of proxy-front points preserved, so the proxy supports coarse ranking and is no substitute for the held-out re-scoring every reported schedule receives. Routed token–expert pairs are an architecture-level work proxy and are identical across the three  $\sum_{\ell} k_{\ell} = 89$  allocations, so uniform  $k = 3$  in Table 2 is a lower-budget endpoint rather than a matched control.